Hao-Ting Wang | Assistant Professor in Data Science

Current affiliation:
Department of Psychiatry
University of British Columbia, Vancouver, British Columbia, Canada

Research area: Computational cognitive neuroscience
Specialisations: Functional magnetic resonance imaging, Neuroinformatics, Biomarker

RESEARCH POSITIONS

Assistant Professor — May 2026 – Present
Department of Psychiatry, University of British Columbia — *Vancouver, BC, Canada*
Other affiliations: Djavad Mowafaghian Centre for Brain Health; UBC Institute of Mental Health

Postdoctoral Researcher — Sept. 2021 – Apr. 2026
CRIUGM — *Montréal, QC, Canada*
Principal Investigator: Prof Lune (Pierre) Bellec
Software for neuroimaging and neurodegeneration biomarker discovery

Research Fellow — Sept. 2019 – Aug. 2021
Sackler Centre for Consciousness Science, University of Sussex — *Brighton, United Kingdom*
Principal Investigators: Prof Hugo Critchley, Prof Sarah Garfinkle
Cognitive processes in psychiatric conditions with neuroimaging and physiology measures

Research Associate — Nov. 2018 – Aug. 2019
University of York — *York, United Kingdom*
Principal Investigator: Prof Jonathan Smallwood

Research Administrator — Oct. 2015 – Oct. 2018
University of York — *York, United Kingdom*
Principal Investigators: Prof Jonathan Smallwood and Prof Elizabeth Jefferies

EDUCATION

PhD in Cognitive Neuroscience and Neuroimaging — Sept. 2015 – Dec. 2018
University of York — *York, United Kingdom*

Master of Research in Psychology — Sept. 2013 – Sept. 2014
University of York — *York, United Kingdom*

BSc in Psychology — Sept. 2009 – June 2013
National Chengchi University — *Taipei, Taiwan*

SCHOLARSHIPS

2024 - 2026	Canadian Neuroanalytics Scholars Program. AB, Canada.	CAD$140,000
2022 - 2024	IVADO Postdoctoral scholarship. QC, Canada.	CAD$70,000
2022	UNIQUE Excellence Scholarship. QC, Canada. (Declined)	CAD$20,000
2019 - 2021	Sackler Foundation Research Fellowship. United Kingdom.	£33,199 per annum

AWARDS

2023	Canadian Trainee Prize Neuro-Irv and Helga Cooper Foundation Open Science Prizes	CAD$5,000
2017	Guarantors of Brain Travel Award Machine Learning Summer School, Tübingen, Germany	£600
2016	The Neuro Bureau Travel Award Brainhack Vienna, Vienna, Austria	USD$500
2014	Department of Psychology Summer Bursary Award University of York , United Kingdom	£1000

TALKS

2026 What do I do in neuroscience and open source software?
Women in Computer Science. University of British Columbia, BC, Canada.
2026 Cognitive Neuroscience in the Foundation Model Era: Lessons in Reproducibility.
Brainhack Vanderbilt. Online.
2025 Towards foundation models for cognitive neurosciences.
Child Mind Institute, New York, USA.
2024 Symposium Speaker. Big Data in Psychiatry: Does It Solve the Reproducibility Problem?
2024 Society of Biological Psychiatry. Austin, TX, USA.
2024 Invited Speaker. Continuous Evaluation of Denoising Strategies in fMRI Using fMRIPrep and Nilearn. COMPASS Neuro-AI seminar series.
2023 Canadian Trainee Prize recipient talk. 5th Annual Neuro Open Science in Action Symposium.
Montreal, QC, Canada.
2023 Symposium Speaker. Hackathon and Early Career Development. OHBM 2023.
Montreal, QC, Canada.
2023 Fantastic open source projects and how to find them. UNIQUE Student Symposium 2023.
Montreal, QC, Canada.
2023 A reproducible benchmark of fMRI denoising strategies in fMRIPrep and Nilearn.
Q-BIN Science Day. Quebec City, QC, Canada.
2022 load_confounds. Neuroimaging in Montreal. Montreal, QC, Canada.
2021 Panel speaker on neuroinformatics at the University of Texas Brainstorms.
2021 Panel speaker on academic career at MAIN 2021.
2021 Panel speaker at SciPy2021 Biology and Neuroscience mini-symposium.
2021 Canonical correlation analysis application in neuroimaging data.
Queen's University, Kingston, Canada.
2019 Recent trend in resting-state functional connectivity. University of Sussex, Brighton, UK.
2019 Data simulation workshop. University of York, York, UK.
2019 Multivariate mapping of functional brain and behaviour. Child Mind Institute, New York, USA.
2018 Small steps to reproducible science. University of York, York, UK.

PROFESSIONAL DEVELOPMENT

Aug. 2019 Neurohackademy, Seattle, USA.
Dec. 2017 Large-scale trends in cortical organization, Leipzig, Germany.
June 2017 Machine Learning Summer School, Tübingen, Germany.
Sep. 2016 Brainhack Vienna, Vienna, Austria.
Feb. 2016 Brainhack@Paris, Paris, France.

PROFESSIONAL SERVICE

Committees

Oct. 2022 – June 2023 Brainhack school organiser. Montreal, QC, Canada.
Oct. 2021 – Sep. 2022 Hackathon Chair, Open Science special interest group, OHBM. Glasgow, UK.
Mar. 2020 – Aug. 2021 ECR representative, Sussex Neuroscience Steering Committee.
University of Sussex, UK.
Jun. 2021 OHBM Sparkle special task force. OHBM, virtual.
Jun. 2021 Live Q & A cohost and general enquiry, OHBM Brainhack. virtual.
Oct. 2018 – Aug. 2019 Member, Open Science Interest Group. University of York, UK.
Oct. 2018 – Aug. 2019 Member, Early Career Researcher forum. University of York, UK.
Mar. 2017 Organizing committee, Brainhack York. York, UK.

Open Source Software

2021 – present	NiLearn	Core developer.
2021	load_confounds	Core developer.
2020	Brainhack book	Contributor and maintainer.
2020 - 2021	Pydra-FSL	Contributor and maintainer.
2019	NiBable	Contributor and reviewer.

Memberships: Organization for Human Brain Mapping (OHBM); Open Science Special Interest Group, OHBM.

Ad-hoc Peer Reviewer: Aperture Neuro, Advances in Methods and Practices in Psychological Science, Brain Imaging and Behavior, Communications Biology, Journal of Open Source Software, IEEE Transactions on Signal and Information Processing over Networks, Imaging Neuroscience, Medical Image Analysis, NeuroImage, Neuroinformatics, Neurobiology of Aging, PCI Registered Reports.

TEACHING EXPERIENCE

Events, Workshops and Summer Schools	**Year**
Brainhack School	2025, 2022
Montreal AI Neuroscience educational workshop	2024, 2022, 2021
OHBM Hackathon	2020

Course

Programming in Neuroimaging 2015 – 2016
University of York

Mentoring

2025	Clara El Khantour	MSc	Université de Montréal
2025	Pierre Bergeret	PhD	Université de Montréal
2025	Seann Wang	Intern	Université de Montréal
2019–2021	Will Strawson	PhD	University of Sussex (with Prof. S. Garfinkle)
2019	Bronte McKeown	MSc	University of York (with Prof. J. Smallwood)
2019	Will Strawson	MSc	University of York (with Prof. J. Smallwood)
2018	Rebecca Lowndes	MSc	University of York (with Prof. J. Smallwood)
2018	Delali Konu	MSc	University of York (with Prof. J. Smallwood)

RECENT PUBLICATIONS

Conference Posters

[1] H.-T. Wang, A. Pichet Binette, P. Sainath, N. Farrugia, S. Villeneuve, and L. Bellec, "A reproducible benchmark of fMRI denoising strategies in fMRIPrep and Nilearn." Bordeaux, France: OHBM, June 2026.

[2] H.-T. Wang, F. Paugam, N. Farrugia, and P. Bellec, "A scaling study of self-supervised auto-regressive modelling of fMRI time series and performance on downstream sex prediction in the UK biobank sample." Boston, MA, USA: Cognitive Computational Neuroscience, August 2024.

[3] H.-T. Wang, S. L. Meisler, H. Sharmarke, N. Clarke, N. Gensollen, C. J. Markiewicz, F. Paugam, B. Thirion, and P. Bellec, "A reproducible benchmark of fMRI denoising strategies in fMRIPrep and Nilearn." Montreal, QC,Canada: OHBM, July 2023.

[4] H.-T. Wang, S. L. Meisler, H. Sharmarke, N. Clarke, N. Gensollen, C. J. Markiewicz, F. Paugam, B. Thirion, and P. Bellec, "A reproducible benchmark of fMRI denoising strategies in fMRIPrep and Nilearn." Pavillon Alphonse-Desjardins, Université Laval, Québec, Quebec: Journée scientifique du RBIQ, May 2023.

[5] H.-T. Wang, N. Clarke, S. L. Meisler, H. Sharmarke, N. Gensollen, C. J. Markiewicz, F. Paugam, B. Thirion, and P. Bellec, "A reproducible benchmark of fMRI denoising strategies in fMRIPrep and Nilearn." Montreal, Canada: IVADO, October 2022.

[6] H.-T. Wang, S. L. Meisler, H. Sharmarke, N. Gensollen, C. J. Markiewicz, F. Paugam, B. Thirion, and P. Bellec, "Impact of confound removal strategies on functional connectivity generated from fmriprep preprocessed data." Glasgow, United Kingdom: OHBM, June 2022.

[7] H.-T. Wang, C. Rae, G. Davies, C. Gould van Praag, A. Seth, H. Critchley, and S. Garfinkel, "Insula hypoactivation is associated with dissociative experiences." Virtual Conference: OHBM, June 2020.

[8] H.-T. Wang, N. S. Ping Ho, D. Bzdok, B. C. Bernhardt, D. S. Margulies, E. Jefferies, and J. Smallwood, "Neurocognitive patterns dissociating semantic processing from executive control are linked to more detailed off-task mental time travel." Seattle, USA: Neurohackademy, August 2019.

[9] H.-T. Wang, N. S. Ping Ho, D. Bzdok, B. C. Bernhardt, D. S. Margulies, E. Jefferies, and J. Smallwood, "Neurocognitive patterns dissociating semantic processing from executive control are linked to more detailed off-task mental time travel." Rome, Italy: OHBM, June 2019.

[10] H.-T. Wang, E. Jefferies, and J. Smallwood, "Inhibition of prior mental content contributes to content representation of on-going thoughts." Montreal, Canada: Resting State Brain Connectivity, Sep. 2018.

[11] H.-T. Wang, D. Bzdok, D. S. Margulies, R. C. Craddock, M. P. Milham, E. Jefferies, and J. Smallwood, "Decomposing self-reports of experience at rest with brain connectivity reveals links to intelligence." Singapore: OHBM, June 2018.

[12] H.-T. Wang, G. L. Poerio, C. Murphy, D. Bzdok, E. Jefferies, and J. Smallwood, "Dimensions of experience: Exploring the heterogeneity of the wandering mind." Amsterdam, Netherlands: 13th International Conference for Cognitive Neuroscience, Aug. 2017.

[13] H.-T. Wang, G. L. Poerio, C. Murphy, D. Bzdok, E. Jefferies, and J. Smallwood, "Dimensions of experience: Exploring the heterogeneity of the wandering mind." Tübingen, Germany: Machine Learning Summer School, Jun. 2017.

[14] H.-T. Wang, D. Bzdok, C. Murphy, D. Vatansever, G. L. Poerio, J. Smallwood, and E. Jefferies, "Component processes and the wandering mind: Links between spontaneous thought contents, task performance and resting state brain connectivity." Vienna, Austria: Resting State Brain Connectivity, Sep. 2016.